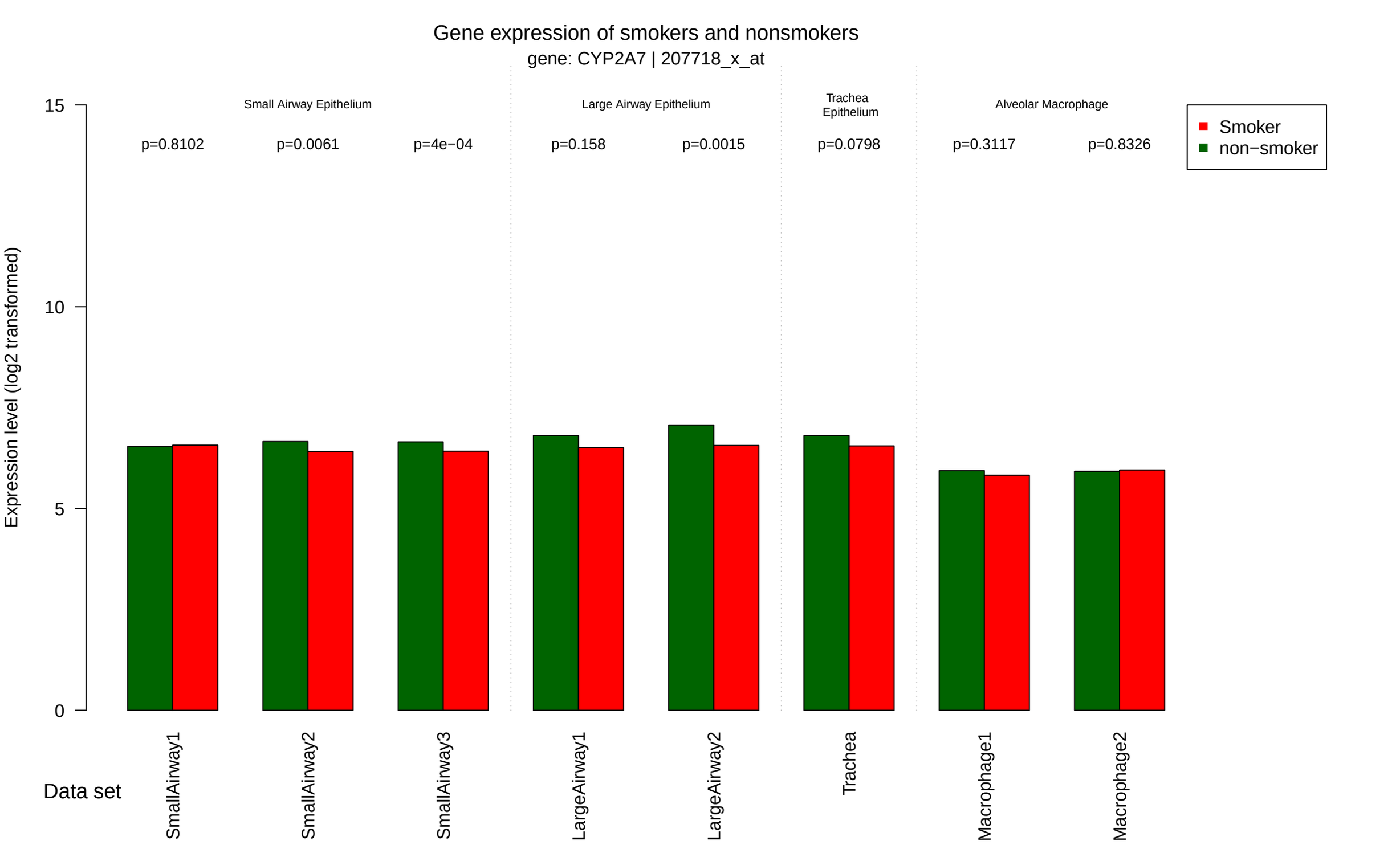
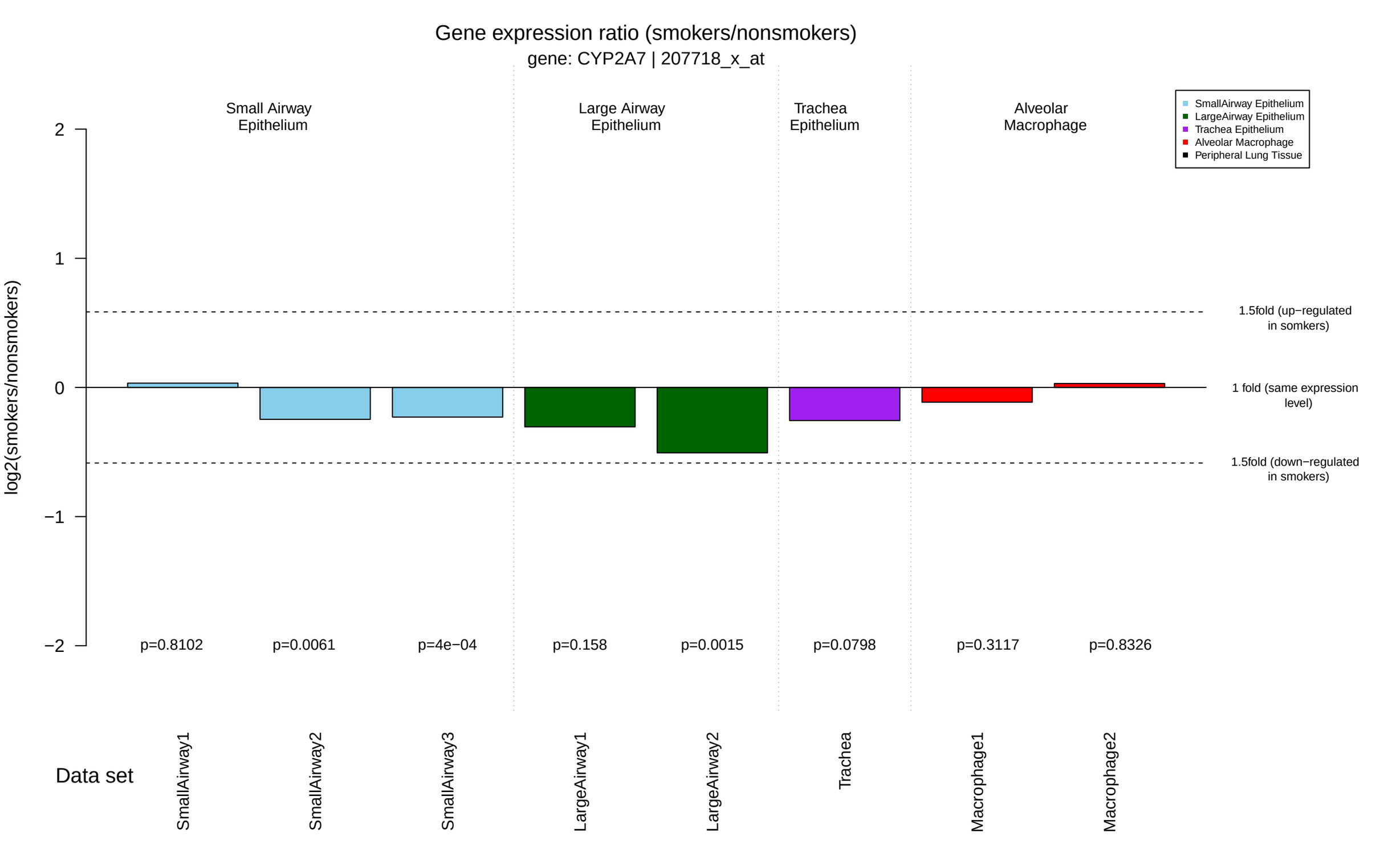
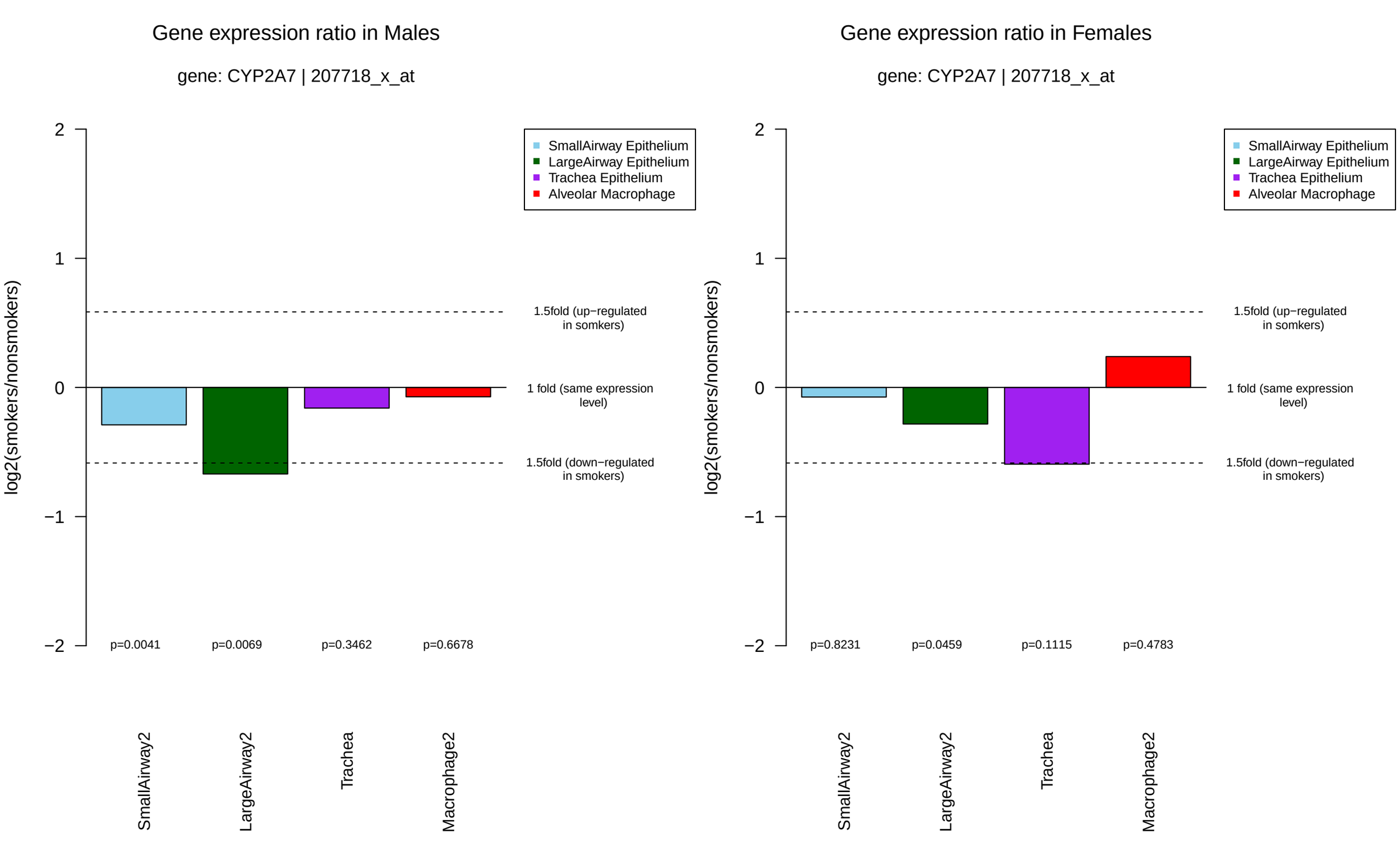
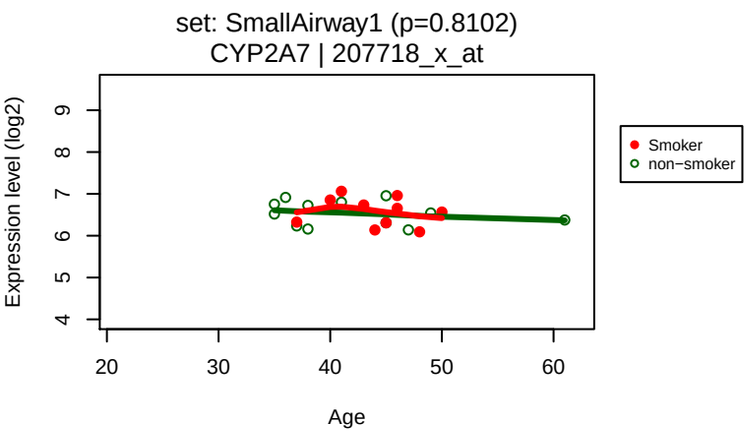
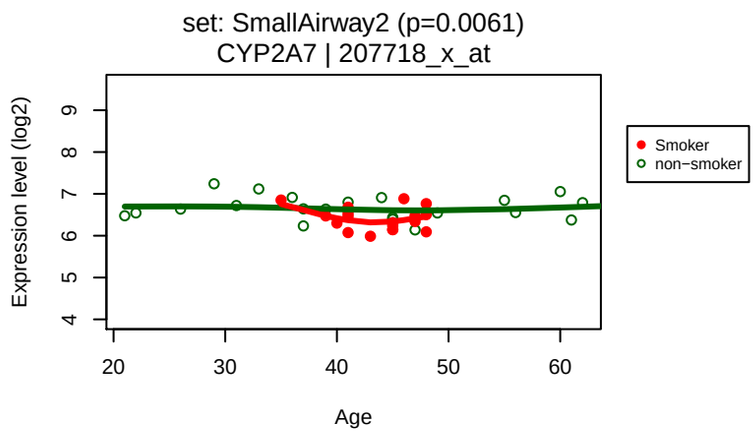
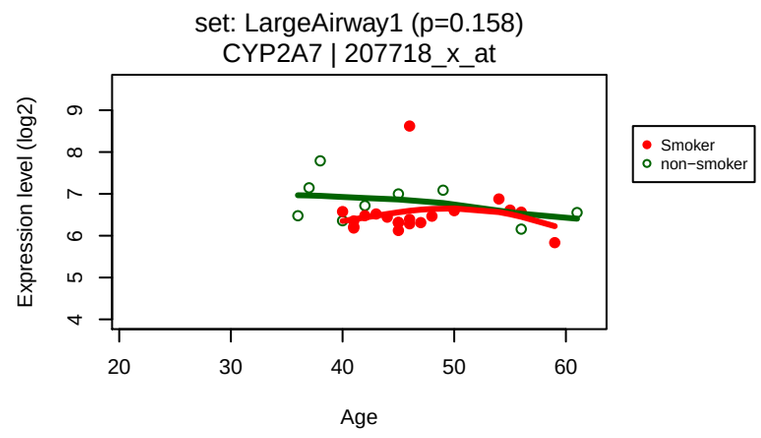

Gene expression of smokers and nonsmokers
gene: CYP2A7 | 207718_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.8102
p=0.0061
p=4e−04
p=0.158
p=0.0015
p=0.0798
p=0.3117
p=0.8326
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

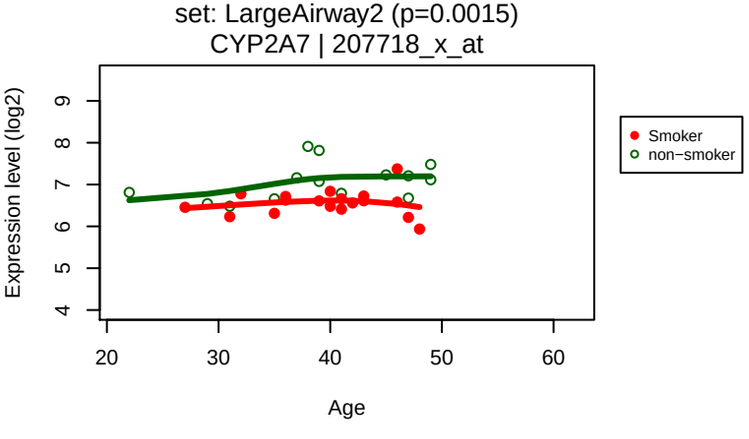

Gene expression ratio (smokers/nonsmokers)
gene: CYP2A7 | 207718_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8102
p=0.0061
p=4e−04
p=0.158
p=0.0015
p=0.0798
p=0.3117
p=0.8326
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

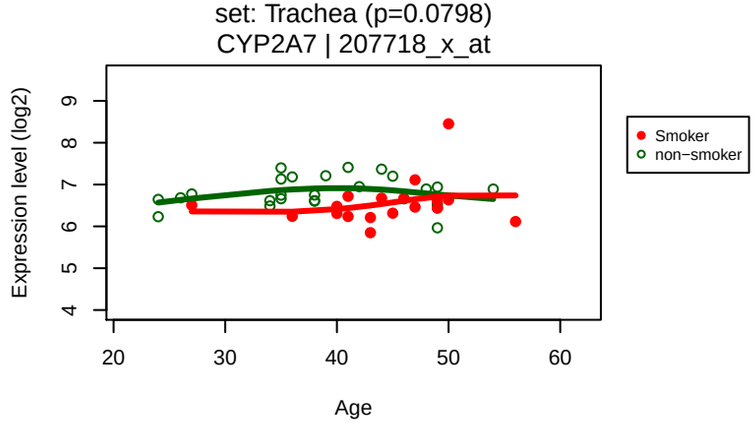

Gene expression ratio in Males
gene: CYP2A7 | 207718_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0041
p=0.0069
p=0.3462
p=0.6678
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: CYP2A7 | 207718_x_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.8231
p=0.0459
p=0.1115
p=0.4783
SmallAirway2
LargeAirway2
Trachea
Macrophage2

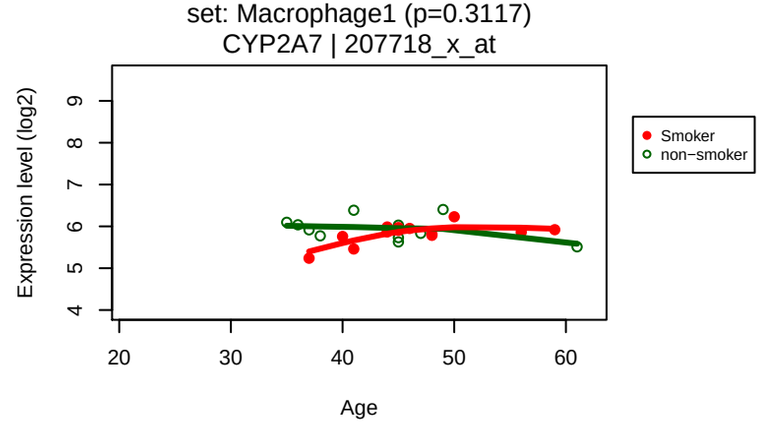

set: SmallAirway1 (p=0.8102)
CYP2A7 | 207718_x_at
Expression level (log2)
Age
Smoker
non−smoker

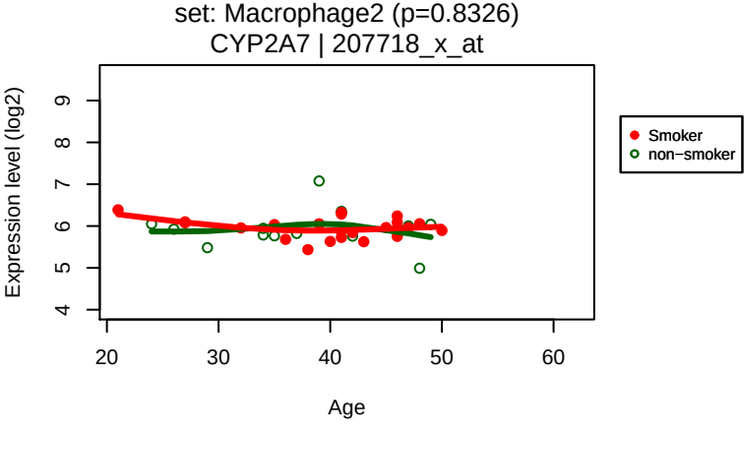

set: SmallAirway2 (p=0.0061)
CYP2A7 | 207718_x_at
Expression level (log2)
Age
Smoker
non−smoker

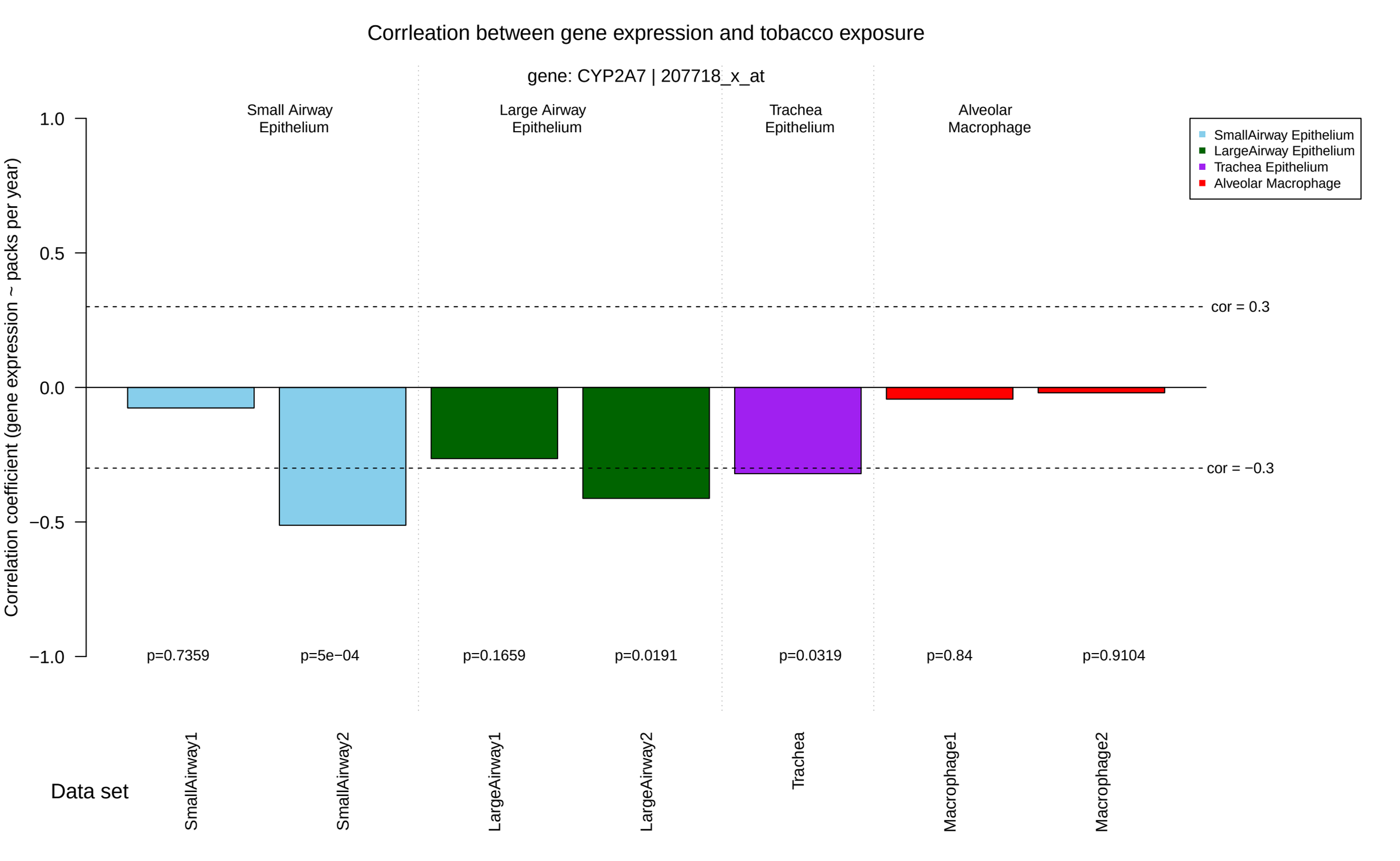

set: LargeAirway1 (p=0.158)
CYP2A7 | 207718_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.0015)
CYP2A7 | 207718_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.0798)
CYP2A7 | 207718_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.3117)
CYP2A7 | 207718_x_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.8326)
CYP2A7 | 207718_x_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: CYP2A7 | 207718_x_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.7359
p=5e−04
p=0.1659
p=0.0191
p=0.0319
p=0.84
p=0.9104
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2